

Wikipedia Empowered Natural Language Interface for Web Search

Haofen Wang¹, S M Mazharul Hoque Chowdhury², and Wei Jin² ✉

¹ College of Design Innovation, Tongji University
carter.whfcarter@gmail.com

² Dept. of Computer Science & Engineering, University of North Texas, USA
{smmazharulhoque.chowdhury, wei.jin}@unt.edu

Abstract. Understanding queries in information retrieval (IR) is crucial. Tasks like query classification or clustering exist but may lack precision. Detailed goal descriptions, like human annotations, are vital for improving query understanding and evaluating document relevance. This paper addresses two problems: (1) how to simplify natural language questions so keyword-based search engines can understand them, and (2) how to make this process lightweight but effective. The first problem is solved by using a context graph to translate questions into keyword queries. The second uses Wikipedia for entity recognition, disambiguation, and relevance feedback. Experiments on 892 TREC questions show a 21.45% overall MRR increase, with 36.41% for "why" questions, compared to Bing, Google, and Yahoo.

Keywords: Information Retrieval, Web Search, Natural Language Interface, Wikipedia

1 Introduction

Since the Web's inception³, text has been the primary information form, with search engines like Bing, Google, and Yahoo using the bag-of-words model for retrieval. While effective for keyword queries, this model struggles with complex natural language queries [1, 2], highlighting the need for natural language interfaces. These interfaces face challenges due to language's flexibility and ambiguity [3], and the high cost of quality labeled datasets in open Web environments complicates the use of statistical learning methods.

This paper addresses two key challenges in applying a natural language interface to Web search: (1) simplifying natural language questions for better keyword-based search engine understanding, and (2) providing lightweight yet effective solutions. The first issue is tackled by creating a context graph that summarizes question context, using entities as keywords to generate a semantic query. To solve the second, Wikipedia is integrated as background knowledge, enabling unsupervised entity recognition, question disambiguation, and relevance feedback, without requiring manual labeling. Our major contributions can be summarized as follows:

³ <http://www.worldwidewebsize.com>

- 1 An algorithm matches phrases in natural language questions to relevant Wikipedia articles, leveraging its knowledge base and avoiding manual labeling.
- 2 The algorithm uses Wikipedia’s articles to create a context graph for disambiguating meanings and generating precise keyword queries.
- 3 Relevance feedback and query expansion techniques enrich the context of natural language questions, selectively enhancing interdependence among entities while minimizing noise from irrelevant feedback.

The paper covers related work (Section 2), workflow (Section 3), context graph (Section 4), query expansion (Section 5), and experimental results (Section 6), followed by the conclusion.

2 Related Work

In this section, we first give a brief introduction of the state of the art of natural language interfaces. We also list our main differences from the existing work. Then we discuss the current status of two major techniques employed in natural language interface, namely named entity recognition and query expansion.

2.1 Natural Language Interface

Recent advancements in natural language understanding have revived Natural Language Interfaces (NLI), evolving from basic keyword search to support complex, context-aware queries [4]. A Quamar et al. provide a comprehensive overview of NLI technologies. However, NLIs face challenges in adaptation, query complexity, and ensuring trust, crucial for widespread enterprise adoption [4]. S M Cheema et al. focus on simplifying the creation of Data Flow Diagrams (DFDs) using NLI, addressing limitations in existing methods [5]. Their approach enables users to generate accurate DFDs but overlooks limitations like web structure dependency and rule-based constraints. K Kolthoff et al. introduce RaWi, a data-driven GUI prototyping approach utilizing NLI to enhance GUI development processes [6]. Their method allows users to retrieve and reuse GUIs via natural language searches, enhancing GUI development, but is limited by reliance on the Rico dataset, reducing prototype diversity. Q N Nguyen et al. investigate user satisfaction differences between chatbot and menu-based interfaces, highlighting potential drawbacks of chatbots in customer service [7]. R C A Iacob et al. provide a survey of neural approaches for Natural Language Interfaces to Databases (NLIDB), highlighting key datasets and top-performing models [8].

2.2 Named Entity Recognition

Named Entity Recognition (NER) is pivotal in extracting structured information from vast textual data [9]. Techniques have been explored in various domains, such as technology identification in patent texts [10] and Judicial Named Entity Recognition (JNER) system [11].

There has also been work on unsupervised entity extraction from the Web [5, 12], typically with a goal of fact extraction. Moreover, efforts have been

made on named entity disambiguation. The traditional method was based on the occurrence statistics under the bag-of words assumption [4]. Recent studies have focused on using Wikipedia [7], [13] to accomplish this task. Innovative approaches in feature fusion, such as the Attentive Multilevel Feature Fusion (AMFF) model, capture multilevel features from various perspectives [14]. With a Context-Aware AMFF model, it outperforms existing baselines, highlighting multilevel feature value.

2.3 Query Expansion

Query Expansion (QE) improves information retrieval by adding relevant terms to enhance precision and recall [12]. Recent studies explore various approaches and their effectiveness. Wenhao Zhu et al. propose the Query Context Expansion-Based Retrieval (QCER) method, which uses pseudo-relevance feedback and dense readers for term selection, significantly enhancing retrieval performance [15]. Liang Wang et al. introduce "query2doc," a straightforward yet effective query expansion method, generating pseudo-documents through large language models to boost retrieval systems like BM25 without requiring fine-tuning [16]. H K Azad et al. present a survey of Query Expansion techniques in Information Retrieval from 1960 to 2017, emphasizing their historical development and significance in enhancing web searches [17]. They propose an innovative approach leveraging Wikipedia and WordNet for QE, resulting in significant improvements in retrieval effectiveness [18]. This work [19] treats Wikipedia entity pages as high-quality pseudo-relevance documents for query expansion, incorporating question context to select terms that align with these entities.

3 Natural Language Interface

Wikipedia extracts entities from questions like "How many miles are there from Hong Kong to Toronto?" (Figure 1), identifying terms such as "miles," "Hong Kong," and "Toronto." A context graph is created and expanded using Wikipedia links, adding new entities like "distance." The final keyword query combines terms from both the initial and enriched graphs for better search results.

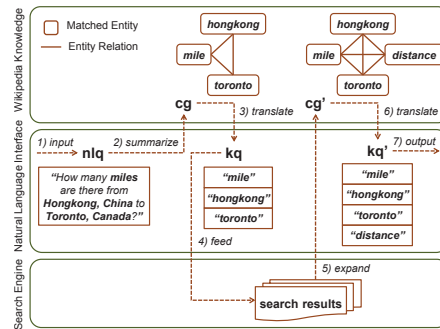


Fig. 1. Workflow of the techniques proposed for answering natural language questions.

4 Context Graph Construction

Traditional Web search engines treat each word in a natural language question as a keyword, causing poor query quality, especially with verbose or ambiguous phrases. For instance, splitting "Hong Kong, China" alters its meaning, leading to irrelevant results. Words like "How" and "are" add minimal value.

4.1 Context Based Question Translation

This paper employs context graphs to enhance the translation of natural language questions into keyword queries, with entities as nodes and relationships as edges. For instance, in "How many miles from Hong Kong to Toronto?", entities like *mile*, *HongKong*, and *Toronto* form the graph. The highest-scoring graph, such as cg_1 in Figure 2, is selected to improve query relevance.

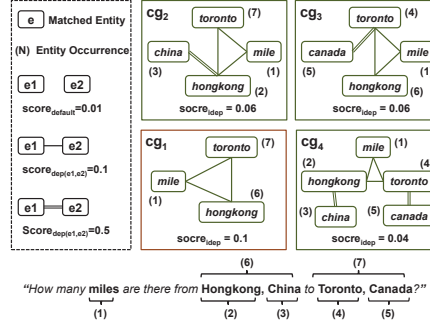


Fig. 2. Four possible context graphs for the example question.

4.2 Construction and Disambiguation

Wikipedia titles and n-grams identify entities in natural language questions via exact matching, as shown in Figure 2 with *miles*, *HongKong*, *China*, *Toronto*, and *Canada*. Redirect pages enhance matching, while disambiguation pages may yield multiple results. Table 1 indicates each entity averages 1.13 redirect and 0.13 disambiguation pages. The question generates combinations like *mile*, *hongkong*, *toronto* and *mile*, *hongkong*, *toronto*, *china*, *canada*. Context graphs are built for these combinations and evaluated by "context energy," with the highest score chosen as the best representation.

First, the dependency set (S_{dep}) of a given entity e is computed which is composed of a set of Wikipedia articles. This set includes the article describing e and the articles e refers to (through page-links). Next, the dependency score ($score_{dep}$) between two entities is measured by the Jaccard similarity between their respective dependency sets, which is given below:

$$S_{dep}(e) = \{article_e\} \cup \{articles \text{ page-linked by } article_e\}$$

$$score_{dep}(e_i, e_j) = \begin{cases} \frac{|S_{dep}(e_i) \cap S_{dep}(e_j)|}{|S_{dep}(e_i) \cup S_{dep}(e_j)|} & \text{if } S_{dep}(e_i) \cap S_{dep}(e_j) \neq \emptyset \\ \text{default value} & \text{otherwise} \end{cases}$$

Here, e_i and e_j denote $entity_i$ and $entity_j$ respectively. Note that a default value is given in the calculation of $score_{dep}$ when $S_{dep}(e_i)$ and $S_{dep}(e_j)$ have no intersection, which is to ensure that each $score_{dep}$ can return a non-zero value for the following calculations.

The interdependence score ($score_{idep}$) of a context graph cg is calculated by a geometric mean of the $score_{dep}(e_i, e_j)$ on each pair of entities contained in the context graph.

$$score_{idep}(cg) = \sqrt[|E|]{\prod_{i=1, j=2, i < j}^E score_{dep}(e_i, e_j) \mid e_i, e_j \in E}$$

Note that E denotes the entity set of a given context graph cg and $|E|$ denotes the total number of entities contained in E . Among the four context graphs in Figure 2, cg_1 has the highest $score_{idep}$ (0.1), making it the clear context representation. We translate it into a semantic keyword query, resulting in "mile," "Hong Kong," and "Toronto."

5 Query Expansion

Here the semantic keyword query (generated in Section 4) is fed into a Web search engine and top N search results are collected for selecting best expansion terms. Three methods are used to choose candidate terms and described as follows.

5.1 Term Selection

1. TFIDF-based method: The TFIDF-based method assesses term importance using document titles and snippets as documents, calculating scores for the top N documents and selecting the top M terms for expansion.

$$tfidf(t, d) = tf(t, d) \cdot \log \frac{|D|}{df(t)}$$

where $tf(t, d)$ denotes the total count of t in a document $d \in D$ and $df(t)$ is the number of documents containing t .

2. The Cluster-Oriented TFIDF method enhances query expansion through hierarchical clustering of documents as TFIDF vectors [20]. The number of clusters is determined by a formula [21], and top distinct terms are selected from updated TFIDF scores to improve quality.

$$d = \langle tfidf(t_1, d), tfidf(t_2, d), \dots, tfidf(t_m, d) \rangle$$

$$cos(d_i, d_j) = \frac{d_i \cdot d_j^T}{|d_i| \cdot |d_j|}$$

Here, d is a document and $t_1, t_2, \dots, t_m$ are its m terms in d . the $tfidf(t_i, d)$ is updated when the document set shifts from all documents to the local cluster containing d .

3. The LDA-based method enhances query expansion using Latent Dirichlet Allocation (LDA) [22] to cluster words into $\sqrt{N/2}$ topics. We extract key phrases from the top N documents and select the top $M/\sqrt{N/2}$ terms from each topic as expansion candidates, improving quality.

5.2 Context Based Expansion

We enhance search results by selecting expansion terms that improve question context. By searching Wikipedia for entities, we add terms to boost the query’s context energy ($score_{idep}$), resulting in terms like "distance," "flight," and "Victoria." Figure 3 shows expanded graphs with $score_{idep}$ values: 0.13 for cg_{1a} , 0.1 for cg_{1b} , and 0.05 for cg_{1c} . We select cg_{1a} for its higher $score_{idep}$ ($score_{idep}(cg_1) = 1.0$). The final query includes "mile," "Hong Kong," "Toronto," and "distance."

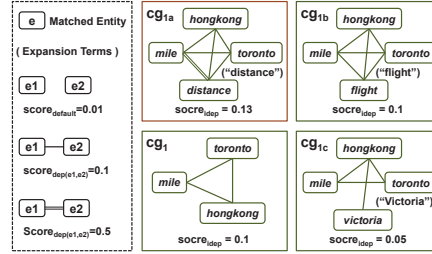


Fig. 3. Context based expansions for the example question.

6 Experiments

In this section, we introduce the experimental settings, describe the evaluation strategy, baseline model, assess question context representation and expansion techniques, and compare performance with our approach.

6.1 Experimental Settings

We downloaded the English Wikipedia dump⁴ and indexed article titles, redirects, and page-links using Lucene⁵. After removing irrelevant pages, Table 1 summarizes key statistics, with only 1.05% of articles lacking page-links.

Table 1. The statistics on fields of Wikipedia.

Field	Total #	Article Coverage	# / Article
Article	3,381K	100.00%	1.00
Redirect	3,808K	44.36%	1.13
Disambiguation	433K	12.53%	0.13
Page Link	81,837K	98.95%	24.20

⁴ <https://dumps.wikimedia.org/>

⁵ <http://lucene.apache.org/>

Table 2. The statistics on TREC question collection.

Category	#Question	Percentage	Example
"How"	5	0.56%	198. "How did Socrates die?"
"How XX"	80	8.97%	334. "How old is the sun?"
"What"	284	31.84%	635. "What is cribbage?"
"What XX"	167	18.72%	471. "What year did Hitler die?"
"When"	67	7.51%	054. "When did Nixon die?"
"Where"	92	10.31%	020. "Where is Inoco based?"
"Who"	165	18.50%	222. "Who is Anubis?"
"Why"	4	0.45%	315. "Why can't ostriches fly?"
"Name a"	20	2.24%	213. "Name a flying mammal."
"Name the"	8	0.90%	890. "Name the highest mountain."

We gathered 892 questions from TREC 8 & 9, categorizing them into 10 types. Table 2 outlines these categories. Our method, though question-word independent, focuses on improving traditional Web search by translating these questions and evaluating performance gains.

In our implementation, we use 1-gram, 2-gram, and 3-gram enumerations to extract entities from natural questions for context representation, setting the default dependency score to 0.0001 (section 4.2). We retrieve search results from Bing, Google, and Yahoo for query expansion, with both document size N and term size M set to 10 as mentioned in Section 5.1. A detailed evaluation of various N values is presented in Section 6.5. Additionally, we use the MALLET package⁶ to perform LDA-based topic modeling on the retrieved documents.

6.2 Evaluation Strategy and Baseline

To determine document relevance, we examine titles and snippets rather than full text. A document is considered relevant if its title or snippet matches the TREC answer template using regular expressions. This approach conserves bandwidth and speeds up evaluation by focusing on concise excerpts.

Table 3. The statistics on derived context graphs (cgs).

Function	#cgs / Question	#Entities / cg	#Relations / cg
Average	7.61	4.73	4.03
Maximum	576	11	31
Minimum	1	1	0

We adopt the Mean Reciprocal Rank (MRR) as the evaluation metric, where Q denotes the question set and $|Q|$ is the total number of questions contained in Q . $rank_i$ denotes the position of the first document relevant to the i -th question. $1/rank_i$ is set to 0 if there are no relevant documents returned. Since people are used to clicking on results on the first page, we retrieved top 10 search results for the MRR evaluation in the following experiments.

$$MRR = \frac{1}{|Q|} \sum_{i=1}^Q \frac{1}{rank_i}$$

The baseline model translates natural language questions word-by-word, treating each word as a keyword. We compare its performance against our translation approach using Bing, Google, and Yahoo across 10 question categories.

⁶ <http://mallet.cs.umass.edu/download.php>

6.3 Effectiveness of Question Context Representation

Table 3 shows statistics for generated context graphs, with an average of 7.61 graphs per question (range: 1 to 576) and an average of 4.73 entities per graph (max: 11, min: 1). The average number of relations per graph is 4.03 (max: 31, some with none). We compare our context-based question translation with the baseline model and an entity-based query translation method, which uses a greedy approach to match Wikipedia article titles to entity occurrences. Figure 5 illustrates MRR comparisons, revealing a 12% MRR increase for EBQT on Yahoo and 6% on Bing and Google compared to the baseline. CBQT, which considers interdependence among entities, achieves a 13% MRR increase on Google and Yahoo, though Bing shows minimal improvement. Table 4 details MRR comparisons 10 question categories, with CBQT outperforming the baseline by an average of 16% across all engines. Notably, "Why"-style questions see a 130% MRR increase, with significance assessed using the paired t-test at $p < 0.05$.

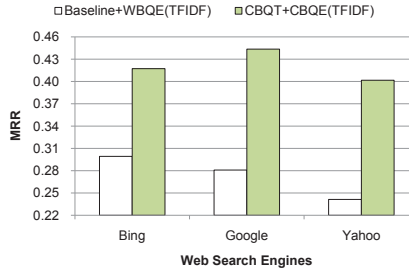


Fig. 4. MRR comparison: weight-based vs. context-based expansion using TFIDF.

6.4 Effectiveness of Query Expansion

Before evaluating methods in Section 5, we compare weight-based query expansion (WBQE) and context-based query expansion (CBQE). WBQE selects the best term from baseline terms, while CBQE uses a TFIDF method (Section 5.1) to choose the top 10 terms from CBQT results. Figure 4 shows that CBQE provides significant MRR gains: 12% on Bing and 16% on Google and Yahoo. Table 5 reveals that CBQT+CBQE outperforms in all categories, achieving over 50% MRR increases in seven categories, including 392% for "Why"-style questions. Overall, CBQT+CBQE leads to a 53% MRR improvement over Baseline+WBQE, with significance at $p < 0.05$.

6.5 Evaluation on Expansion Methods and Document Sizes

Figure 6 shows MRR variations across query expansion methods and document sizes. The baseline ("N = 0") uses CBQT without expansion, while the LDA method consistently outperforms CBQT across all search engines.

Leveraging the context graph improves MRR, while TFIDF and Cluster-oriented TFIDF expansions decrease performance by selecting irrelevant terms. The LDA-based method excels with top-20 to top-40 document sizes. Optimal document sizes (N values) yield MRR gains of 3% on Bing, 4% on Google, and 6% on Yahoo compared to CBQT.

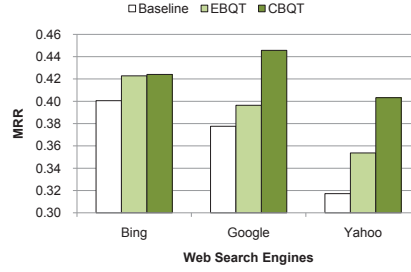


Fig. 5. MRR comparisons: baseline, entity-based, and context-based translations.

Table 4. MRR comparisons of baseline vs. context-based translation (CBQT) on Bing, Google, and Yahoo across 10 question categories, with paired t-test at $p < 0.05$.

Category	Baseline			CBQT			Average Increase
	Bing	Google	Yahoo	Bing	Google	Yahoo	
"How"	0.41	0.27	0.50	0.27	0.51	0.44	3.41%
"How XX"	0.12	0.16	0.13	0.18	0.13	0.17	16.83%
"What"	0.40	0.37	0.28	0.41	0.45	0.41	19.81%
"What XX"	0.43	0.40	0.33	0.46	0.48	0.43	17.29%
"When"	0.34	0.29	0.34	0.39	0.42	0.33	17.16%
"Where"	0.49	0.39	0.39	0.49	0.47	0.48	13.27%
"Who"	0.48	0.51	0.40	0.52	0.56	0.47	11.42%
"Why"	0.12	0.05	0.10	0.25	0.11	0.25	129.96%
"Name a"	0.29	0.38	0.31	0.39	0.44	0.35	19.76%
"Name the"	0.51	0.55	0.49	0.52	0.52	0.71	13.48%
ALL	0.40	0.38	0.32	0.42	0.45	0.40	16.22%

Table 5. Table 6 shows MRR across 10 categories, highlighting a 21.45% average increase for CBQT+CBQE over the baseline, with significance ($p < 0.05$).

Category	Baseline+WBQE(TFIDF)			CBQT+CBQE(TFIDF)			Average Increase
	Bing	Google	Yahoo	Bing	Google	Yahoo	
"How"	0.00	0.12	0.24	0.27	0.51	0.44	240.19%
"How XX"	0.16	0.11	0.07	0.16	0.14	0.19	41.55%
"What"	0.29	0.26	0.24	0.40	0.43	0.38	53.27%
"What XX"	0.27	0.35	0.22	0.44	0.46	0.43	56.41%
"When"	0.26	0.27	0.25	0.37	0.42	0.32	42.86%
"Where"	0.42	0.29	0.28	0.52	0.50	0.49	53.96%
"Who"	0.37	0.34	0.32	0.51	0.58	0.49	55.25%
"Why"	0.13	0.00	0.00	0.25	0.11	0.25	391.67%
"Name a"	0.30	0.32	0.28	0.39	0.44	0.39	36.44%
"Name the"	0.31	0.23	0.39	0.52	0.52	0.63	79.28%
ALL	0.30	0.28	0.24	0.42	0.44	0.40	53.69%

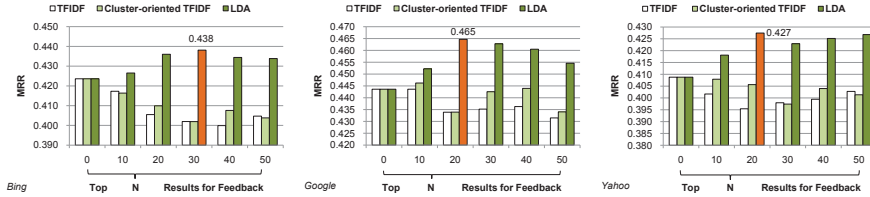


Fig. 6. MRR comparison on Bing, Google, and Yahoo for query expansion methods.

6.6 Summary of Experimental Results

Figure 7 compares MRR between the baseline, CBQT, and CBQE. CBQT improves MRR by 6% on Bing, 18% on Google, and 27% on Yahoo, while CBQE

enhances results with an additional 3% on Bing, 4% on Google, and 6% on Yahoo. Table 6 shows that CBQT+CBQE achieves a 21.45% average MRR increase in all categories, with over 25% improvement in four categories. Paired t-tests confirm the statistical significance of these improvements ($p < 0.05$).

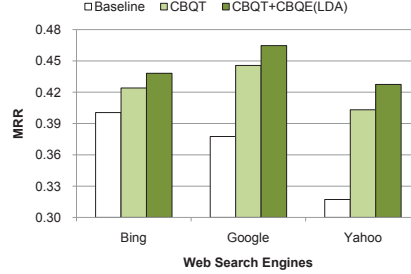


Fig. 7. MRR comparisons of baseline, context-based question translation, and context-based query expansion using LDA ($N = 30$ for Bing, $N = 20$ for Google and Yahoo).

Table 6. Comparison of MRR between baseline and CBQE using LDA-based expansion ($N = 30$ for Bing, $N = 20$ for Google and Yahoo), with significance at $p < 0.05$.

Category	Baseline			CBQT+CBQE(LDA)			Average Increase
	Bing	Google	Yahoo	Bing	Google	Yahoo	
"How"	0.41	0.27	0.50	0.27	0.51	0.44	3.41%
"How XX"	0.12	0.16	0.13	0.17	0.12	0.17	10.13%
"What"	0.40	0.37	0.28	0.41	0.46	0.42	21.52%
"What XX"	0.43	0.40	0.33	0.50	0.50	0.45	25.15%
"When"	0.34	0.29	0.34	0.34	0.42	0.30	9.16%
"Where"	0.49	0.39	0.39	0.58	0.53	0.52	28.11%
"Who"	0.48	0.51	0.40	0.54	0.60	0.53	20.02%
"Why"	0.12	0.05	0.10	0.00	0.11	0.25	36.41%
"Name a"	0.29	0.38	0.31	0.36	0.47	0.40	25.94%
"Name the"	0.51	0.55	0.49	0.52	0.58	0.78	21.56%
ALL	0.40	0.38	0.32	0.44	0.46	0.43	21.45%

7 Conclusion

This paper presents a method for integrating natural language into web search using Wikipedia for entity identification and context graphs for disambiguation. It reformulates queries into semantic keywords, improving search performance through relevance feedback. Future work aims to optimize parameters and scale for larger question-answering systems.

References

1. Jie Zou, Aixin Sun, Cheng Long, Mohammad Aliannejadi, and Evangelos Kanoulas. Asking clarifying questions: To benefit or to disturb users in web search? *Information Processing Management*, 60(2):103176, 2023.
2. In-Ho Kang and Gil-Chang Kim. Query type classification for web document retrieval. In *SIGIR*, 2003.
3. Rongwei Cen, Yiqun Liu, Min Zhang, Liyun Ru, and Shaoping Ma. Automatic search engine performance evaluation with the wisdom of crowds. In *AIRS*, 2009.
4. Abdul Quamar, Vasilis Efthymiou, Chuan Lei, and Fatma Özcan. Natural language interfaces to data. *Found. Trends Databases*, 11(4):319–414, 2022.

5. Sehrish Munawar Cheema, Saman Tariq, and Ivan Miguel Pires. A natural language interface for automatic generation of data flow diagram using web extraction techniques. *Journal of King Saud University-Computer and Information Sciences*, 35(2):626–640, 2023.
6. Kristian Kolthoff, Christian Bartelt, and Simone Paolo Ponzetto. Data-driven prototyping via natural-language-based gui retrieval. *Automated Software Engineering*, 30(1):13, 2023.
7. Quynh N Nguyen, Anna Sidorova, and Russell Torres. User interactions with chatbot interfaces vs. menu-based interfaces: An empirical study. *Computers in Human Behavior*, 128:107093, 2022.
8. Radu Cristian Alexandru Iacob, Florin Brad, Elena-Simona Apostol, Ciprian-Octavian Truică, Ionel Alexandru Hosu, and Traian Rebedea. Neural approaches for natural language interfaces to databases: A survey. In *proceedings of the 28th International Conference on Computational Linguistics*, pages 381–395, 2020.
9. Zara Nasar, Syed Waqar Jaffry, and Muhammad Kamran Malik. Named entity recognition and relation extraction: State-of-the-art. *ACM Computing Surveys (CSUR)*, 54(1):1–39, 2021.
10. Giovanni Puccetti, Vito Giordano, Irene Spada, Filippo Chiarello, and Gualtiero Fantoni. Technology identification from patent texts: A novel named entity recognition method. *Technological Forecasting and Social Change*, 186:122160, 2023.
11. Hu Zhang, Jiayu Guo, Yujie Wang, Zhen Zhang, and Hansen Zhao. Judicial nested named entity recognition method with mrc framework. *International Journal of Cognitive Computing in Engineering*, 4:118–126, 2023.
12. Enwei Zhu and Jinpeng Li. Boundary smoothing for named entity recognition. *arXiv preprint arXiv:2204.12031*, 2022.
13. Georgios Peikos, Daria Alexander, Gabriella Pasi, and Arjen P de Vries. Investigating the impact of query representation on medical information retrieval. In *European Conference on Information Retrieval*, pages 512–521. Springer, 2023.
14. Zhiwei Yang, Jing Ma, Hechang Chen, Jiawei Zhang, and Yi Chang. Context-aware attentive multilevel feature fusion for named entity recognition. *IEEE transactions on neural networks and learning systems*, 2022.
15. Wenhao Zhu, Xiaoyu Zhang, Liang Ye, and Qiuhong Zhai. Query context expansion for open-domain question answering. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 2023.
16. Liang Wang, Nan Yang, and Furu Wei. Query2doc: Query expansion with large language models. *arXiv preprint arXiv:2303.07678*, 2023.
17. Hiteshwar Kumar Azad and Akshay Deepak. Query expansion techniques for information retrieval: a survey. *Information Processing & Management*, 56(5):1698–1735, 2019.
18. Hiteshwar Kumar Azad and Akshay Deepak. A new approach for query expansion using wikipedia and wordnet. *Information sciences*, 492:147–163, 2019.
19. Massimo Esposito, Emanuele Damiano, Aniello Minutolo, Giuseppe De Pietro, and Hamido Fujita. Hybrid query expansion using lexical resources and word embeddings for sentence retrieval in question answering. *Information Sciences*, 514:88–105, 2020.
20. Ying Zhao and George Karypis. Evaluation of hierarchical clustering algorithms for document datasets. In *CIKM*, 2002.
21. K.V. Mardia, J.T. Kent, J.M. Bibby, et al. *Multivariate analysis*. Academic press London, 1979.
22. David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent dirichlet allocation. *Journal of Machine Learning Research*, 2003.